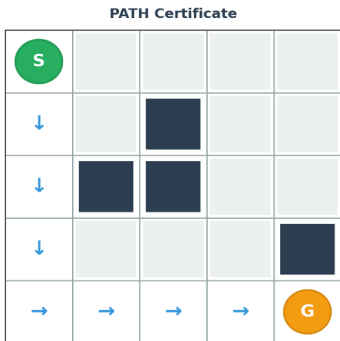


Send the Proof, Not the Data

✓ CertTalk: Send the certificate

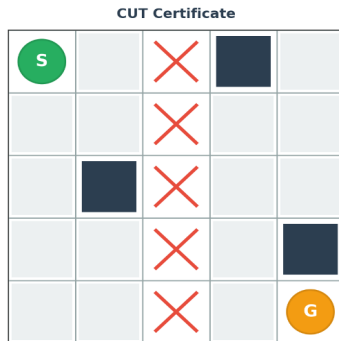
If reachable → PATH_CERT



"Here's the route"

~12 bytes

If blocked → CUT_CERT



"This wall blocks us"

~20 bytes

What is a Certificate?

Certificate = Claim + Artifact + Attestation

Claim: "Path exists" or "Unreachable" • **Artifact:** RLE-encoded path/cut • **Attestation:** Witness bits

To verify: Certificate + Union map only. **Send-All** sends coords → rebuild + solve. **CertTalk** sends proof → direct check.

Breaking Epistemic Deadlock

How agents certify with partial knowledge

The Problem

A proposes cut with 6 cells. B can only verify 3 of them:



What B sees

Without attribution:

- B must NACK (can't verify 3 cells)
- A sends PROBE
- Deadlock: 5+ messages

The Solution

Witness bits (1 per cell) encode WHO vouches:



Who attests each cell

B's acceptance rule:

Accept cell if: **(I see blocked) OR (bit=0)**

- All 6 cells pass ✓
- Fast path: 3 messages

Cost: 1 byte (6 bits) • Benefit: Eliminates probe round (~200 bytes)

What We'll Compare

Systems under test

Send-All

(baseline)

"Dump raw data"

A → All obstacle coords

B → All obstacle coords

B → Rebuild map

B → Run pathfinder

B → Answer

Opaque - need solver

CertTalk

(this work)

"Send proof"

A → Certificate proposal

B → Verify locally

B → Certificate ✓

Final artifact: PATH or CUT proof

Interpretable - visual proof

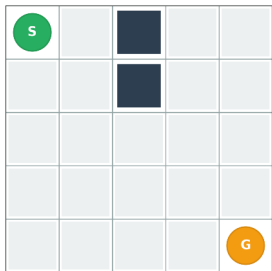
Preliminary 10×10 results:

Send-All: ~350B (100%) • CertTalk: ~470B (100%) • +34% overhead for proof

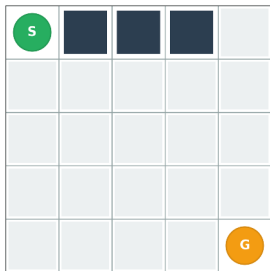
+ Other baselines (exploration strategies) tested on same 1,000 instances

How We'll Test

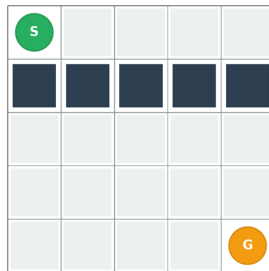
We'll generate 1,000 cached grids (4×4 to 22×22, mixed density)



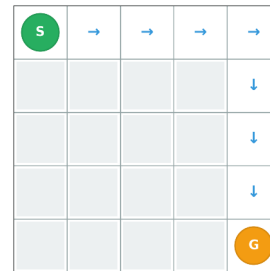
Path exists



Path exists



Blocked barrier



Trivial path

Constraints (same for all):

- Max 64 messages
- Max 256 bytes per packet
- Max 3,072 bytes total

What we'll measure & why:

- ✓ **Correctness** (oracle accepts) → Does it work?
- ✓ **Efficiency** (bytes, messages) → Practical overhead?
- ✓ **Interpretability** (human verification time) → Auditability feasible?

Timeline & Deliverables

Oct 26 → Dec 6 (6 weeks)

Week 1-2

Week 3-4

Week 5-6

**Core
Implementation**

Protocol + Encodings

Baselines & Dataset

1,000 grids + Test harness

**Experiments &
Analysis**

Full runs + Paper

Final Deliverables:

Code (GitHub)

Dataset (1K grids)

Paper (Arxiv)

Presentation

Risks & Mitigation

Anticipated challenges and our strategies

⚠️ Computational Scalability

Simulation caps at 22×22 grids (1,000 seeds max)

✓ Mitigation

Focus on 4×4 to 22×22 range, vary density, document limits

⚠️ Timeline Pressure

Dec 6 deadline: experiments + analysis + paper

✓ Mitigation

Test other baselines only once Send-All vs CertTalk work rigorously